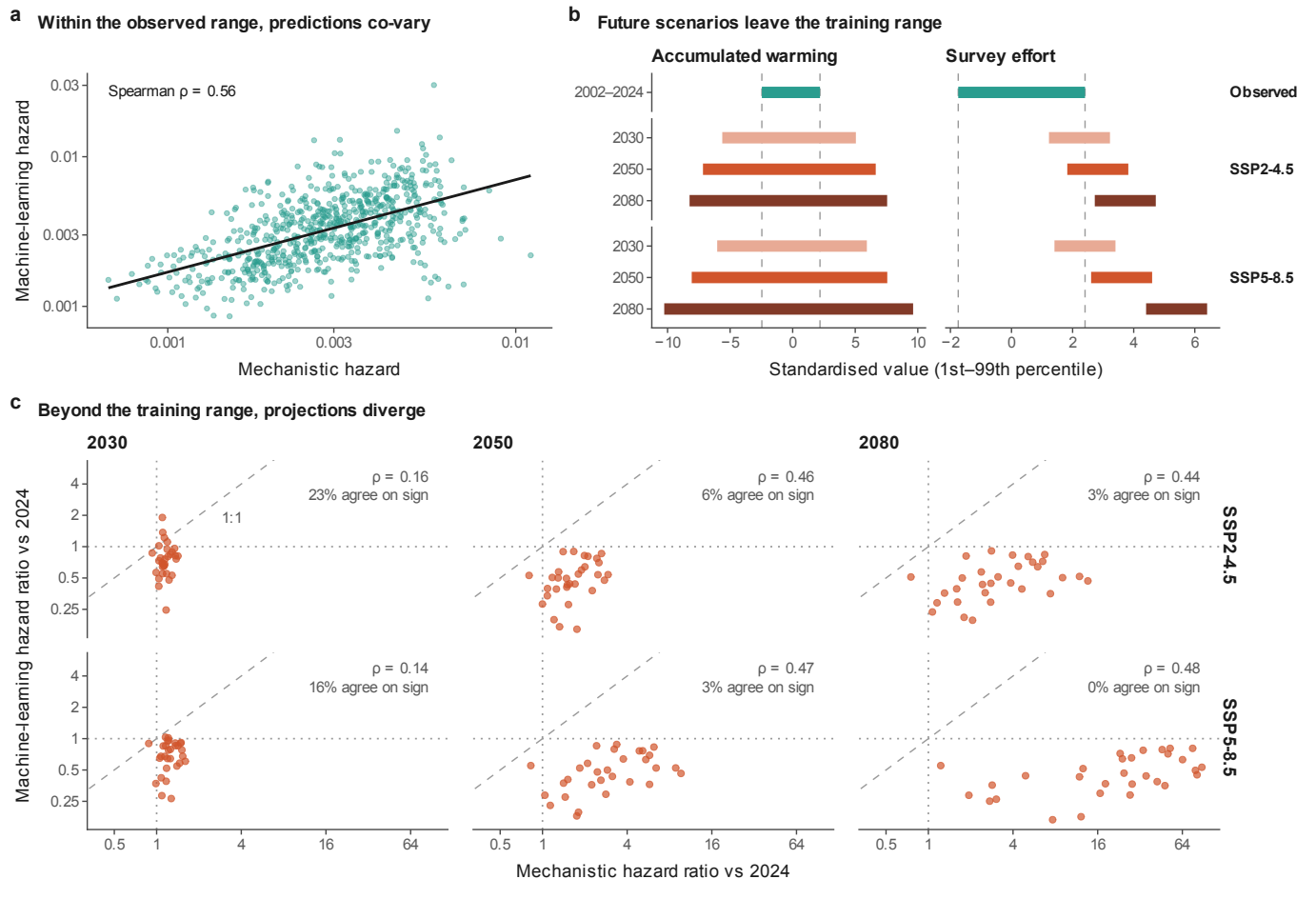

a
Within the observed range, predictions co-vary
Spearman ρ = 0.56
Machine-learning hazard
Mechanistic hazard
0.03
0.01
0.003
0.001
b
Future scenarios leave the training range
Accumulated warming
Survey effort
2002–2024
2030
2050
2080
Observed
SSP2-4.5
SSP5-8.5
Standardised value (1st–99th percentile)
c
Beyond the training range, projections diverge
2030
2050
2080
ρ = 0.16
23% agree on sign
1:1
ρ = 0.46
6% agree on sign
ρ = 0.44
3% agree on sign
ρ = 0.14
16% agree on sign
ρ = 0.47
3% agree on sign
ρ = 0.48
0% agree on sign
Machine-learning hazard ratio vs 2024
Mechanistic hazard ratio vs 2024